

PROBABILITY & STATISTICS FOR LIFE SCIENCES

Independence, Bernoulli Trials, and the Binomial Distribution

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Purpose and standing assumptions

This article is a reference for explaining the existing Chapter 5 slides. It continues from the properties of expectation and variance, develops independence through centered deviations, and then builds the Bernoulli and binomial models from individual trials. The final step is the derivation of the binomial mean and variance.

All random variables in this article are **discrete and have finite second moments**. In particular, every expectation, variance, and covariance used below exists. Write $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$. The notation $\text{Var}(X)$ means the same thing as $V(X)$ in the course slides.

The main thread is a question: *When may we add expectations, when may we add variances, and how do those facts help us understand a count of successes?* Keep the distinction between an individual outcome, a random count, and the probability of that count visible throughout.

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1 Independence: learning one value leaves the other distribution unchanged

Two random variables X and Y are independent when every pair of values has a joint probability that factors into its two marginal probabilities:

Definition of independence

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y) \quad \text{for every pair } (x, y).$$

The word *every* matters. One factorization for one pair does not establish independence of the random variables.

Here $\mathbb{P}(X = x, Y = y)$ means the probability that *both* equalities hold. A marginal probability, such as $\mathbb{P}(Y = y)$, describes one variable without conditioning on the other. If $\mathbb{P}(X = x) > 0$, independence gives

$$\begin{aligned} \mathbb{P}(Y = y \mid X = x) &= \frac{\mathbb{P}(X = x, Y = y)}{\mathbb{P}(X = x)} && \text{definition of conditional probability} \\ &= \frac{\mathbb{P}(X = x)\mathbb{P}(Y = y)}{\mathbb{P}(X = x)} && \text{independence} \\ &= \mathbb{P}(Y = y). \end{aligned}$$

Thus, after learning $X = x$, the probabilities assigned to the possible values of Y stay the same. The corresponding statement also holds with X and Y reversed whenever the conditioning probability is positive.

Conversely, if the conditional distribution of Y is unchanged for every x with positive probability, multiplying by $\mathbb{P}(X = x)$ recovers the joint-probability definition. Values of x with probability zero cause no problem: all their joint probabilities are zero as well.

An information statement, not a causal statement

Independence describes a probability distribution. It says that learning one value does not change the distribution of the other. It does not, by itself, establish whether one physical process causes or influences another.

2 Linearity, products, and covariance

2.1 Expectation is linear without independence

For constants a, b, c , we always have

$$\mathbb{E}[aX + bY + c] = a\mathbb{E}[X] + b\mathbb{E}[Y] + c.$$

The two random variables may be independent, strongly related, or even the same random variable. None of those possibilities changes this rule. For example, $\mathbb{E}[X + X] = 2\mathbb{E}[X]$ is still linearity.

One way to see why dependence does not obstruct addition is to use the joint probability mass function.

Summing over all possible values,

$$\begin{aligned}
 \mathbb{E}[X + Y] &= \sum_x \sum_y (x + y) \mathbb{P}(X = x, Y = y) \\
 &= \sum_x x \underbrace{\sum_y \mathbb{P}(X = x, Y = y)}_{\mathbb{P}(X=x)} + \sum_y y \underbrace{\sum_x \mathbb{P}(X = x, Y = y)}_{\mathbb{P}(Y=y)} \\
 &= \mathbb{E}[X] + \mathbb{E}[Y].
 \end{aligned}$$

We only added the joint probabilities over the other variable's possible values. We did not factor the joint probability into a product.

2.2 Independence lets an expected product factor

For a product, a different step is needed. If X and Y are independent,

$$\begin{aligned}
 \mathbb{E}[XY] &= \sum_x \sum_y xy \mathbb{P}(X = x, Y = y) \\
 &= \sum_x \sum_y xy \mathbb{P}(X = x) \mathbb{P}(Y = y) && \text{independence is used here} \\
 &= \sum_x x \mathbb{P}(X = x) \left(\sum_y y \mathbb{P}(Y = y) \right) \\
 &= \left(\sum_x x \mathbb{P}(X = x) \right) \left(\sum_y y \mathbb{P}(Y = y) \right) \\
 &= \mathbb{E}[X] \mathbb{E}[Y].
 \end{aligned}$$

The inner sum no longer depends on x , so it can be taken outside the outer sum. Finite second moments ensure that these sums can be rearranged: in particular, $2|XY| \leq X^2 + Y^2$ gives $\mathbb{E}[|XY|] < \infty$.

Two rules with different requirements

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] \quad \text{always,}$$

$$\mathbb{E}[XY] = \mathbb{E}[X] \mathbb{E}[Y] \quad \text{if } X \text{ and } Y \text{ are independent.}$$

Independence is sufficient for the product rule. We will soon see that it is not necessary.

2.3 Covariance records the expected product of deviations

The centered values $X - \mu_X$ and $Y - \mu_Y$ tell us how far each variable lies above or below its own mean. Their product is positive when their signs agree, negative when the signs differ, and zero when either deviation is zero. Covariance averages these products:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)].$$

Expanding the expression gives a useful computational formula:

$$\begin{aligned}
 \text{Cov}(X, Y) &= \mathbb{E}[XY - \mu_Y X - \mu_X Y + \mu_X \mu_Y] \\
 &= \mathbb{E}[XY] - \mu_Y \mathbb{E}[X] - \mu_X \mathbb{E}[Y] + \mu_X \mu_Y \\
 &= \mathbb{E}[XY] - \mu_X \mu_Y - \mu_X \mu_Y + \mu_X \mu_Y \\
 &= \boxed{\mathbb{E}[XY] - \mathbb{E}[X] \mathbb{E}[Y]}.
 \end{aligned}$$

The means are fixed numbers, so linearity permits pulling them outside expectations. Independence was not used in this derivation.

3 Why variances add: inspect the cross term

3.1 Center first, then expand

Set

$$A = X - \mathbb{E}[X], \quad B = Y - \mathbb{E}[Y].$$

These centered deviations have mean zero:

$$\mathbb{E}[A] = \mathbb{E}[X] - \mathbb{E}[X] = 0, \quad \mathbb{E}[B] = 0.$$

The expectations of their squares recover the variances, $\mathbb{E}[A^2] = \text{Var}(X)$ and $\mathbb{E}[B^2] = \text{Var}(Y)$, while their expected product is $\mathbb{E}[AB] = \text{Cov}(X, Y)$.

Linearity gives $\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$, so the deviation of the sum from its mean is exactly $A + B$. Now expand the square before taking averages:

$$\begin{aligned} \text{Var}(X + Y) &= \mathbb{E}\left[\left((X + Y) - \mathbb{E}[X + Y]\right)^2\right] \\ &= \mathbb{E}[(A + B)^2] \\ &= \mathbb{E}[A^2 + 2AB + B^2] \\ &= \mathbb{E}[A^2] + 2\mathbb{E}[AB] + \mathbb{E}[B^2] \\ &= \boxed{\text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)}. \end{aligned}$$

The extra term is not a technical afterthought: it measures whether the two deviations tend to reinforce or oppose one another.

For a difference, the centered deviation is $A - B$ instead:

$$\begin{aligned} \text{Var}(X - Y) &= \mathbb{E}[(A - B)^2] \\ &= \mathbb{E}[A^2 - 2AB + B^2] \\ &= \boxed{\text{Var}(X) + \text{Var}(Y) - 2\text{Cov}(X, Y)}. \end{aligned}$$

The individual squared terms remain positive; it is the *cross term* whose sign changes. We do not subtract $\text{Var}(Y)$.

3.2 The precise job of independence

If X and Y are independent, their centered versions A and B are also independent: subtracting a fixed number simply relabels the possible values. The product rule therefore gives

$$\mathbb{E}[AB] = \mathbb{E}[A]\mathbb{E}[B] = 0 \cdot 0 = 0.$$

This is why the cross term disappears. It does not require either distribution to be symmetric.

Independent sums and differences

When X and Y are independent,

$$\boxed{\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)}, \quad \boxed{\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)}.$$

The same formulas also hold whenever their covariance is zero, even if they are dependent.

3.3 Exactly what is equivalent to what?

For two variables with finite second moments, the identities we have derived show the exact equivalences

Equivalent statements for this pair of variables

$$\begin{aligned} \text{Cov}(X, Y) &= 0 \\ \Updownarrow \\ \mathbb{E}[XY] &= \mathbb{E}[X]\mathbb{E}[Y] \\ \Updownarrow \\ \text{Var}(X + Y) &= \text{Var}(X) + \text{Var}(Y) \\ \Updownarrow \\ \text{Var}(X - Y) &= \text{Var}(X) + \text{Var}(Y). \end{aligned}$$

Independence implies all four statements. None of them, by itself, implies independence.

There is an important distinction between a *centered* product and a product of arbitrary random variables. For the centered A, B above, $\mathbb{E}[AB] = \text{Cov}(X, Y)$. To keep this separate from centering, use fresh names U, V for arbitrary random variables. In general,

$$\mathbb{E}[UV] = \text{Cov}(U, V) + \mathbb{E}[U]\mathbb{E}[V].$$

Thus $\mathbb{E}[UV] = 0$ holds exactly when the covariance and the product of the means cancel:

$$\text{Cov}(U, V) = -\mathbb{E}[U]\mathbb{E}[V].$$

Under independence, the covariance is zero, so $\mathbb{E}[UV] = 0$ holds exactly when at least one of $\mathbb{E}[U]$ and $\mathbb{E}[V]$ is zero. Without independence, even having *both* means zero is not enough.

4 Examples and the many-variable rule

4.1 Zero covariance can hide dependence

Let X be uniform on $\{-1, 0, 1\}$ and define $Y = X^2$. There are three possible joint outcomes, each with probability $1/3$:

X	$Y = X^2$	XY	Probability
-1	1	-1	1/3
0	0	0	1/3
1	1	1	1/3

Read each expectation as a weighted average of its column:

$$\begin{aligned}\mathbb{E}[X] &= (-1)\frac{1}{3} + 0\frac{1}{3} + 1\frac{1}{3} = 0, \\ \mathbb{E}[Y] &= 1\frac{1}{3} + 0\frac{1}{3} + 1\frac{1}{3} = \frac{2}{3}, \\ \mathbb{E}[XY] &= (-1)\frac{1}{3} + 0\frac{1}{3} + 1\frac{1}{3} = 0.\end{aligned}$$

Therefore

$$\text{Cov}(X, Y) = 0 - 0\left(\frac{2}{3}\right) = 0.$$

But knowing X determines Y completely. For example,

$$\mathbb{P}(Y = 0 \mid X = 0) = 1 \quad \text{whereas} \quad \mathbb{P}(Y = 0) = \frac{1}{3}.$$

The conditional distribution changes, so the variables are dependent. This one example also disproves the claim that $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$ implies independence. That equality checks one weighted average; independence checks the entire joint distribution.

4.2 Symmetry and zero means do not remove dependence

Let A take -1 and 1 with equal probabilities and let $B = A$. Both distributions are symmetric about zero, and $\mathbb{E}[A] = \mathbb{E}[B] = 0$. Nevertheless,

$$AB = A^2 = 1 \quad \text{on every outcome,} \quad \mathbb{E}[AB] = 1.$$

The deviations always have the same sign. Symmetry in each distribution does not make the two variables independent and does not force their expected product to vanish.

4.3 Several variables: each pair contributes a cross term

For $X_1, \dots, X_n$, expand the square of the sum of their centered deviations. Each squared term appears once; each distinct pair appears twice. Consequently,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j).$$

The notation $i < j$ means that every pair is counted once. If the variables are independent, all pairwise covariances are zero and the variances add. In fact, *pairwise independence* is sufficient for this variance calculation, and pairwise zero covariance is sufficient as well. For three or more variables, variance additivity for the total only says that the *sum* of the cross covariances is zero; individual nonzero covariances can cancel.

4.4 Two independent copies versus twice the same variable

Suppose X_1 and X_2 are independent copies, each with variance σ^2 . Then

$$\text{Var}(X_1 + X_2) = \sigma^2 + \sigma^2 = 2\sigma^2.$$

If we double one variable instead, we double every deviation from its mean:

$$\text{Var}(2X) = \mathbb{E}[(2X - 2\mathbb{E}[X])^2] = 4\mathbb{E}[(X - \mathbb{E}[X])^2] = 4\sigma^2.$$

Equivalently, $\text{Cov}(X, X) = \text{Var}(X) = \sigma^2$, so adding X to itself includes the extra $2\sigma^2$ cross term.

Relationship	What we may conclude
Any X, Y	Expectations add; variances include $2 \text{Cov}(X, Y)$.
Independent X, Y	Expected products factor; covariance is zero; variances add.
$\text{Cov}(X, Y) = 0$	Expected products factor and variances add; independence is not established.
Symmetric distributions with zero means	No general conclusion that $\mathbb{E}[XY] = 0$.
Independent equal-variance copies	Variance of their sum is $2\sigma^2$.
Twice the same variable	Variance is $4\sigma^2$.

5 One trial: the Bernoulli distribution

5.1 Code the outcome as zero or one

A Bernoulli random variable records whether one specified event occurs. We call the event a “success” and code it as 1; otherwise the outcome is called a “failure” and coded as 0. These are neutral labels, not judgments about whether an outcome is desirable.

Write $p \in [0, 1]$ for the success probability and $q = 1 - p$ for the failure probability. Then $X \sim \text{Bernoulli}(p)$ has probability mass function

Outcome	Value x	$\mathbb{P}(X = x)$
Failure	0	$q = 1 - p$
Success	1	p

The expectation is the probability-weighted average of these numerical values:

$$\mathbb{E}[X] = 0 \cdot q + 1 \cdot p = p.$$

The mean equals p because of the zero–one coding.

5.2 Variance from squared deviations

Since the mean is p , the two deviations are $0 - p$ and $1 - p = q$. We square each deviation and weight it by the probability of its outcome:

Value	Deviation	Squared deviation	Weight	Contribution
0	$-p$	p^2	q	$p^2 q$
1	$1 - p = q$	q^2	p	$q^2 p$

Thus

$$\begin{aligned}
 \text{Var}(X) &= (0 - p)^2 q + (1 - p)^2 p \\
 &= p^2 q + q^2 p \\
 &= pq(p + q) && \text{factor out the common factor } pq \\
 &= \boxed{pq = p(1 - p)} && \text{because } p + q = 1.
 \end{aligned}$$

The squares and the probability weights have different jobs: p^2 is the squared deviation for outcome 0, while q is its probability.

5.3 A second derivation using the zero-one coding

For either possible outcome, $X^2 = X$: both $0^2 = 0$ and $1^2 = 1$. Consequently $\mathbb{E}[X^2] = \mathbb{E}[X] = p$. The variance shortcut gives

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = p - p^2 = p(1 - p).$$

Notice that $\mathbb{E}[X^2] = p$, whereas $(\mathbb{E}[X])^2 = p^2$. Taking a square before averaging and squaring an average are different operations.

5.4 When is a Bernoulli outcome most variable?

At $p = 0$, the outcome is certainly 0; at $p = 1$, it is certainly 1. In either case, the variance is zero because there is no uncertainty in the outcome. To find the largest variance, regard

$$V(p) = p(1 - p) = p - p^2, \quad 0 \leq p \leq 1,$$

as a function of p . Its derivatives are

$$V'(p) = 1 - 2p, \quad V''(p) = -2.$$

The derivative is zero at $p = 1/2$, and the negative second derivative shows that this is a maximum. Comparing with the zero-variance endpoints,

$$\boxed{\max_{0 \leq p \leq 1} p(1 - p) = \frac{1}{4}, \quad \text{attained at } p = \frac{1}{2}.}$$

A trial is most uncertain when its two outcomes are equally likely. The variance is $1/4$; the corresponding standard deviation is $1/2$.

6 Repeated trials: a binomial count

Fix a number of trials n . Each trial has two outcomes, the same success probability p , and the trials are *mutually independent*. Introduce an indicator for each trial:

$$X_i = \begin{cases} 1, & \text{trial } i \text{ is a success,} \\ 0, & \text{trial } i \text{ is a failure.} \end{cases} \quad X_i \sim \text{Bernoulli}(p).$$

The total number of successes is

$$\boxed{S = X_1 + \cdots + X_n \sim \text{Bin}(n, p).}$$

The symbol S here is a random *count*, taking values $0, 1, \dots, n$. It is not a probability. We use the upright letters S and F for the success and failure symbols in a sequence.

The model assumptions each matter: a fixed number of trials, independent outcomes, and a common success probability. Pairwise independence was enough to remove covariance terms, but the binomial model uses mutual independence to multiply probabilities along an entire outcome sequence.

6.1 Three coin flips: exactly two heads

Suppose three independent flips each have head probability p . There are three distinct sequences with exactly two heads:

Sequence	Probability
HHT	$p \cdot p \cdot q = p^2q$
HTH	$p \cdot q \cdot p = p^2q$
THH	$q \cdot p \cdot p = p^2q$

Independence lets us multiply within each sequence. The three sequences are disjoint outcomes, so we add their probabilities:

$$\mathbb{P}(\text{exactly two heads}) = p^2q + p^2q + p^2q = 3p^2q.$$

For a fair coin, this is $3(1/2)^2(1/2) = 3/8$. The factor 3 counts which positions contain the two heads.

7 Count the positions, then attach their probabilities

7.1 Five slots and exactly two successes: all ten arrangements

First ignore probabilities and ask only where the successes can go. Group the possible strings by the location of their first success:

First success in slot	Strings	Number
1	SSFFF, SFSFF, SFFSF, SFFFS	4
2	FSSFF, FSFSF, FSFFS	3
3	FFSSF, FFSFS	2
4	FFFSS	1
Total		$4 + 3 + 2 + 1 = 10$

Once the two success positions are chosen, every other slot must contain a failure. There is no further choice to make for those remaining slots.

7.2 Why dividing by a factorial removes overcounting

Temporarily distinguish the two successes by calling them S_A and S_B . There are five choices for the position of S_A and four remaining choices for the position of S_B :

$$5 \cdot 4 = 20 \quad \text{placements of the labeled successes.}$$

But the actual experiment records only success or failure, not an A or B label. For example,

$$S_A F S_B F F \quad \text{and} \quad S_B F S_A F F$$

both become the same string **SFSFF** when the labels are erased. Every success-position pattern was counted exactly $2! = 2$ times. Therefore the number of distinct success/failure strings is

$$\binom{5}{2} = \frac{5 \cdot 4}{2!} = 10.$$

The division does not discard possible outcomes. It corrects how often our temporary labeling counted each actual outcome.

For k labeled successes in n slots, the corresponding ordered count is

$$n(n-1) \cdots (n-k+1).$$

There are $k!$ ways to permute the labels among any fixed set of k success positions. Dividing by this common overcount gives

$$\begin{aligned} \binom{n}{k} &= \frac{n(n-1) \cdots (n-k+1)}{k!} \\ &= \boxed{\frac{n!}{k!(n-k)!}}. \end{aligned}$$

The factor $(n-k)!$ enters because $n(n-1) \cdots (n-k+1) = n!/(n-k)!$; it is not an additional choice of failure locations after the successes have been placed. At the endpoints, there is one all-failure string and one all-success string, so $\binom{n}{0} = \binom{n}{n} = 1$, consistent with $0! = 1$.

7.3 Match one probability factor to each letter

Return to **SFSFF**, this time with independent trials:

slot	1	2	3	4	5
outcome	S	F	S	F	F
probability factor	p	q	p	q	q

Thus the probability of this *particular ordered sequence* is

$$p \cdot q \cdot p \cdot q \cdot q = p^2 q^3.$$

Every one of the ten strings has two successes and three failures, so every one has that same probability. Adding the ten disjoint outcomes,

$$\mathbb{P}(S = 2) = 10p^2 q^3 = \binom{5}{2} p^2 q^3.$$

7.4 The general binomial probability mass function

The same reasoning works for any $k \in \{0, 1, \dots, n\}$:

1. A specified sequence with k successes and $n-k$ failures has probability $p^k q^{n-k}$, by independence.
2. There are $\binom{n}{k}$ such sequences.
3. They are disjoint and all have the same probability, so their probabilities add.

Therefore, for $0 < p < 1$,

Binomial probability mass function

$$\mathbb{P}(S = k) = \binom{n}{k} p^k q^{n-k}, \quad k = 0, 1, \dots, n, \quad q = 1 - p.$$

It is zero for other values of k . The coefficient counts the sequences; the powers give the probability of one sequence.

What is being multiplied?

The factor $p^k q^{n-k}$ comes from k individual success factors and $n - k$ individual *failure* factors in one specified sequence. It is not the product of two count-event probabilities such as “the probability of k successes” and “the probability of $n - k$ successes.” Exactly k successes already forces $n - k$ failures.

8 Why the binomial probabilities sum to one

8.1 A five-trial fair example

When $n = 5$ and $p = q = 1/2$, each of the $2^5 = 32$ success/failure strings has probability $1/32$. Grouping them by the number of successes gives

Success count k	0	1	2	3	4	5
Number of strings $\binom{5}{k}$	1	5	10	10	5	1
$\mathbb{P}(S = k)$	1/32	5/32	10/32	10/32	5/32	1/32

Every probability is nonnegative, and

$$\frac{1 + 5 + 10 + 10 + 5 + 1}{32} = \frac{32}{32} = 1.$$

These groups exhaust the possible strings and do not overlap.

8.2 Pascal's triangle counts the same groups

The rows below list $\binom{n}{k}$ as k runs from 0 to n :

Number of trials n	Counts in increasing order of k					
0	1					
1	1	1				
2	1	2	1			
3	1	3	3	1		
4	1	4	6	4	1	
5	1	5	10	10	5	1

To count length- n strings with k successes, look at their last slot. If the last outcome is a success, the first $n - 1$ slots must contain $k - 1$ successes. If it is a failure, the first $n - 1$ slots must contain k successes. The two cases are disjoint, giving, for $1 \leq k \leq n - 1$,

$$\boxed{\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}}.$$

For instance, $\binom{5}{2} = \binom{4}{1} + \binom{4}{2} = 4 + 6 = 10$. The endpoint counts are both 1. The $n = 0$ row records the one empty sequence, which has zero successes.

8.3 The binomial theorem is the algebra of these choices

Expand a product of five identical factors:

$$(q + p)^5 = (q + p)(q + p)(q + p)(q + p)(q + p).$$

To obtain a term, choose either q or p from each parenthesis. A term with two choices of p and three choices of q has value p^2q^3 ; there are ten ways to make those choices. Combining equal terms gives

$$(q + p)^5 = q^5 + 5pq^4 + 10p^2q^3 + 10p^3q^2 + 5p^4q + p^5.$$

More generally, choosing the k parentheses that contribute p gives

$$(q + p)^n = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k}.$$

Since $p + q = 1$, the sum of the binomial probabilities is

$$\boxed{\sum_{k=0}^n \mathbb{P}(S = k) = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k} = 1^n = 1.}$$

For $0 < p < 1$, every term is nonnegative, so these numbers form a valid probability mass function.

8.4 Endpoint probabilities and the distinction from expectation

At $p = 0$, every trial fails: $\mathbb{P}(S = 0) = 1$, with all other count probabilities zero. At $p = 1$, every trial succeeds: $\mathbb{P}(S = n) = 1$, with all other probabilities zero. State these distributions directly rather than asking students to interpret 0^0 in the formula. If $n = 0$ is allowed, the count is zero with probability one for any p .

Normalization and expectation are different sums:

$$\underbrace{\sum_{k=0}^n \mathbb{P}(S = k) = 1}_{\text{add the probabilities}} \quad \text{versus} \quad \underbrace{\mathbb{E}[S] = \sum_{k=0}^n k \mathbb{P}(S = k)}_{\text{weight each value by its probability}}.$$

Proving that the probabilities sum to one has not yet computed the average number of successes. We will derive that average from indicators after examining the cereal example.

9 Read a binomial histogram and evaluate a decision rule

9.1 Each bar represents a possible count

For $X \sim \text{Bin}(n, p)$, the horizontal axis lists a count k , and the height of its bar is $\mathbb{P}(X = k)$. A group of bars represents an event: for example, $X \leq 4$ includes the bars at 0, 1, 2, 3, 4, and its probability is the sum of their heights. These are discrete masses; the gaps between the integer counts carry no probability.

For a fixed n , small p concentrates probability near small counts; large p concentrates it near large counts. The histogram need not be symmetric. A mean marker is a probability-weighted balance point, not necessarily a possible count and not a decision threshold.

9.2 The cereal company's claim and the chosen rule

Suppose a cereal company claims that *at least* 15% of its boxes contain prizes. Let p be the actual probability that a sampled box contains a prize. Treat the outcomes for the 50 sampled boxes as independent, each with the same probability p . Let X count the boxes with prizes, so $X \sim \text{Bin}(50, p)$.

The claim is $p \geq 0.15$, not necessarily $p = 0.15$. Consider the rule chosen before observing the sample:

Decision rule

$$X \leq 4 \implies \text{reject the claim,}$$

$$X \geq 5 \implies \text{do not reject the claim.}$$

“Do not reject” means this rule has not supplied enough evidence against the claim. It does not establish that the claim is true.

9.3 False rejection at the boundary of a true claim

First take the boundary value $p = 0.15$. Write $\mathbb{P}_{0.15}$ to indicate that probabilities are computed under this particular true prize rate. Then

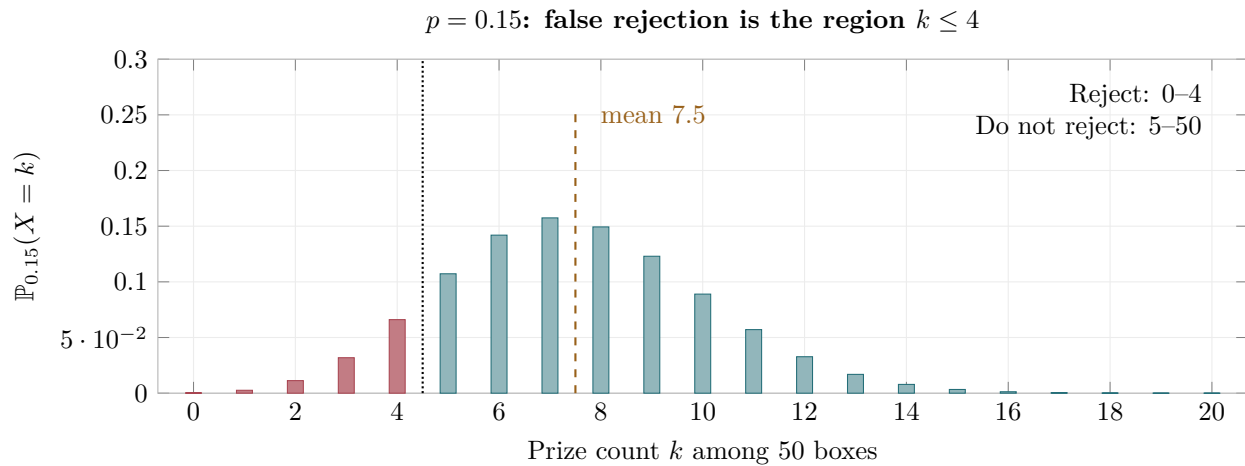
$$\begin{aligned} \mathbb{P}_{0.15}(\text{reject}) &= \mathbb{P}_{0.15}(X \leq 4) \\ &= \sum_{k=0}^4 \mathbb{P}_{0.15}(X = k) \\ &= \sum_{k=0}^4 \binom{50}{k} (0.15)^k (0.85)^{50-k} \\ &\approx \boxed{0.1121 \text{ or } 11.21\%}. \end{aligned}$$

At a true prize rate of 15%, repeated samples of 50 boxes would trigger this false rejection about 11.21% of the time.

Why use the boundary? As the true prize probability increases, low counts become less likely. One way to picture this is to turn some failure outcomes into successes while leaving all existing successes in place. The total count can only increase, so the event “at most four successes” cannot become more likely. Therefore

$$\mathbb{P}_p(X \leq 4) \leq \mathbb{P}_{0.15}(X \leq 4) \quad \text{for every } p \geq 0.15.$$

The value 0.1121 is the *maximum* false-rejection probability over all prize rates consistent with the claim. The actual probability may be smaller if the true rate exceeds 15%.



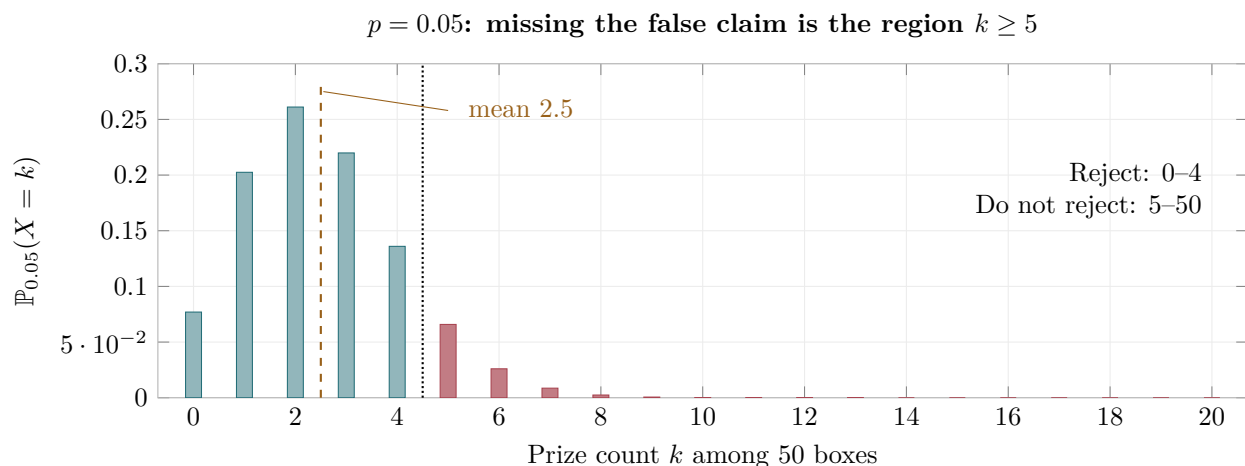
The red bars at 0–4 total 0.1121. The dotted divider sits between the two decisions; the dashed gold line marks the mean, derived below. For readability the plot shows 0–20 on the full support 0–50. The omitted probability $\mathbb{P}_{0.15}(X \geq 21)$ is about 3.90×10^{-6} and belongs to the do-not-reject region.

9.4 Missing a false claim when the actual rate is five percent

Now change the assumed truth to $p = 0.05$. The sampling rule remains the same, but $X \sim \text{Bin}(50, 0.05)$. The company claim is false, and the error of interest is *not* rejecting it:

$$\begin{aligned}
 \mathbb{P}_{0.05}(\text{do not reject}) &= \mathbb{P}_{0.05}(X > 4) \\
 &= 1 - \mathbb{P}_{0.05}(X \leq 4) \\
 &= 1 - \sum_{k=0}^4 \binom{50}{k} (0.05)^k (0.95)^{50-k} \\
 &\approx \boxed{0.1036 \text{ or } 10.36\%}.
 \end{aligned}$$

Because X is an integer, $X > 4$ means $X \geq 5$. At a true rate of 5%, this rule misses the false claim about 10.36% of the time.



The red region now begins at 5. Its total probability, including the unplots counts 21–50, is 0.1036. The omitted probability is about 7.79×10^{-15} . Both histograms use the same horizontal and vertical scales so their shapes and decision regions can be compared directly.

The mean here is 2.5, but “above the mean” is not the same event as “do not reject.” Counts 3 and 4 are above 2.5 and still lead to rejection. The chosen cutoff, not the mean, determines the decision.

9.5 Two errors, computed under two different truths

Assumed true rate	Error made by the rule	Probability
$p = 0.15$	Reject a true claim: $X \leq 4$	0.1121
$p = 0.05$	Do not reject a false claim: $X \geq 5$	0.1036

These are different error types under different values of the true parameter. They are not complementary probabilities, and we do not add them to obtain one overall error probability without specifying how often the different true situations arise.

What these numbers do and do not answer

The first number is a probability of a decision *given a true prize rate*. It is not the probability that the company’s claim is true or false after seeing a sample, nor the probability that a particular conclusion is wrong. Also, 11.21% alone does not establish that a sample of 50 boxes is insufficient. Adequacy depends on the sample size, the cutoff, the departures we want to detect, and which errors we are willing to tolerate.

For example, make rejection harder by changing the cutoff to $X \leq 3$. This removes the $X = 4$ bar from the rejection region. It reduces false rejection when the claim is true, but it also adds $X = 4$ to the region where a false claim is missed:

Rejection rule	False rejection at $p = 0.15$	Miss at $p = 0.05$
$X \leq 4$	0.1121	0.1036
$X \leq 3$	0.0460	0.2396

The sample size and the cutoff together determine the performance of the rule. This tradeoff is the intuitive bridge to later work on statistical testing; the binomial distribution already lets us quantify it.

10 Finish with the binomial mean and variance

10.1 Mean: add the expectations of the indicators

Return to the trial indicators. For $S \sim \text{Bin}(n, p)$,

$$S = X_1 + \cdots + X_n, \quad \mathbb{E}[X_i] = p, \quad \text{Var}(X_i) = pq.$$

Instead of directly evaluating the weighted sum $\sum_{k=0}^n k \binom{n}{k} p^k q^{n-k}$, use linearity:

$$\begin{aligned}
 \mathbb{E}[S] &= \mathbb{E}[X_1 + \cdots + X_n] \\
 &= \mathbb{E}[X_1] + \cdots + \mathbb{E}[X_n] \\
 &= \underbrace{p + \cdots + p}_{n \text{ terms}} \\
 &= np.
 \end{aligned}$$

This addition of expectations does not require independence. Even if the trial outcomes were dependent, a sum of n indicators each having success probability p would still have mean np ; its distribution would not necessarily be binomial.

10.2 Variance: independence removes the cross covariances

For the binomial model the trial indicators are independent. Therefore

$$\begin{aligned}\text{Var}(S) &= \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{i < j} \text{Cov}(X_i, X_j) \\ &= \sum_{i=1}^n pq + 0 && \text{independence makes the covariances zero} \\ &= \boxed{npq = np(1-p)}.\end{aligned}$$

This is where independence matters in the moment calculation. Pairwise zero covariance would suffice for the variance identity, but independence is supplied by our binomial assumptions.

Binomial moments

$$S \sim \text{Bin}(n, p) \implies \mathbb{E}[S] = np, \quad \text{Var}(S) = np(1-p).$$

The mean adds because expectation is linear. The variances add because the independent indicators have zero cross covariances.

For the two cereal distributions,

Distribution	Mean prize count	Variance
$\text{Bin}(50, 0.15)$	$50(0.15) = 7.5$	$50(0.15)(0.85) = 6.375$
$\text{Bin}(50, 0.05)$	$50(0.05) = 2.5$	$50(0.05)(0.95) = 2.375$

A sample cannot contain 7.5 prize-bearing boxes. The fractional mean describes the long-run average count across repeated samples of 50 boxes under the model; it is neither a possible single-sample count nor a guarantee about the next sample. The variance describes the spread of those counts around their mean and is measured in squared count units.

Closing check: which addition used independence?

In $\mathbb{E}[X_1 + \dots + X_n] = \sum_i \mathbb{E}[X_i]$, independence was not needed. In $\text{Var}(X_1 + \dots + X_n) = \sum_i \text{Var}(X_i)$, we used independence to remove the covariance terms. More precisely, zero cross covariances are sufficient. This is the distinction that turns the one-trial Bernoulli calculations into the binomial mean np and variance npq .